Draft Warehouse Prototype Review

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Changes from August 25 version shown in green.

On July 30, 2026, the Summary of Major Comments section was updated to follow the structure of four groups that later reviews contained. All changes reflect putting the comments into groups unless otherwise noted in square brackets [].

Warehouse Scorecard Spreadsheet

Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Pre-1978 metal building u-factors should be higher and depend on climate (WHSC13)
- Infiltration rate for 2008-2019 and 2019+ seem wrong (WHSC14)

The following are the highest priority comments:

- Loading dock infiltration and the thermal effect of goods should be added (WHSC7)
- Cooling by using large exhaust fans not represented and higher exhaust requirements may be justified (WHSC16)
- Fine storage should have humidity control (WHSC18)
- Infiltration decrease during occupied hours is not realistic (WHSC29)
- Bulk storage areas should be cooled (WHSC31)

The following comments represent other important issues found:

- The floor and wall area are some of most important parameters (WHSC4, WHSC5)
- Add use of destratification fans (WHSC17)
- Radiant heating systems common (WHSC20)
- Older vintage interior/exterior lighting power levels seem unrealistic (WHSC25, WHSC26)
- Equipment internal load assumptions should be reexamined (WHSC27, WHSC28)
- Electric battery forklifts required (WHC33)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- What is the area above the office zone used for? (WHSC6-Prototype) [added to this list]
- How was an inner surface of gypsum board determined? (WHSC10-Envelope) [added to this list]

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Interior Lighting
- Ext Lighting
- Equipment
- Schedules
- Energy Usage
- Weights
- PV
- EV Charger

The comments on each tab are labeled using the following format WHSCx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells referenced in the comment.

WHSC1-Notes (n/a)

Sometimes, the specific benefits of code changes are not captured in a set of prototype models because they do not include a building with a specific feature. This has become an

issue for PNNL, where a great fraction of the addenda for 90.1 with energy savings cannot be quantified using the models. See Table 4 of PNNL's "ANSI/ASHRAE/IES Standard 90.1-2022: Energy Savings Analysis - February 2024," where many addenda with energy savings were not included in the quantitative analysis and instead were just discussed qualitatively. 23 of the 39 were not captured quantitatively. In some ways, general prototypes cause this exclusion of understanding impacts, rather than custom prototypes that look at a particular measure. I'm not suggesting a change, but it is a warning that a good system of prototypes still does not answer all questions. Note this is the same comment as PPD1 in my single-family residential prototype review.

WHSC2-Notes (n/a)

The impact weight of a measure can be misleading when using a standardized set of prototype models. The scenario is when a measure is part of a technology that is represented in only one or a few prototype buildings in the set. In that case, the impact of the measure may be more indicative of the weight of that building rather than the weight of the technology in general. Take air-cooled chillers, for example. Perhaps they appear in only one prototype model for commercial buildings. An improvement measure for air-cooled chilled water plants could be significant, but the weight of one building may show that it has a small impact. This is because the weight of one building is based on the building type weighting, which may have nothing to do with the weighting of the air-cooled chiller market. In fact, it would be impossible to weigh the market size for every technology across a standardized set of building prototypes since the data usually does not exist. This subtlety makes it important to understand the frequency of appearance of a given technology in the set of prototypes, which is very important to disclose when assessing the overall impact of a measure. Note this is the same comment as PPD2 in my single-family residential prototype review.

WHSC3-Notes (A1:C21)

Additional data sources could be added to the list shown in the spreadsheet:

- Northwest Energy Efficiency Alliance – Commercial Building Stock Assessment 2019
  https://neea.org/commercial-building-stock-assessment/
- Commercial Buildings Energy Consumption Survey by DOE EIA
  https://www.eia.gov/consumption/commercial/

- Natural Resources Canada – Survey of Commercial and Institutional Energy Use 2019 https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/scieu/2019/tables.cfm
- Advanced Energy Design Guide for small warehouses and self-storage buildings: 30% energy savings. ASHRAE 2008. https://www.ashrae.org/technical-resources/aedgs/30-percent-aedg-free-download
- Technical Support Document – The Development of the Advanced Energy Design Guide for Small Warehouse and Self-Storage Buildings https://www.pnnl.gov/main/publications/external/technical_reports/pnnl-17056.pdf
- Industrial Warehousing Study by Southern California Association of Governments https://scag.ca.gov/post/industrial-warehousing-study
- HVAC Applications. 2023 ASHRAE Handbook. Chapter 3 Section 3
- https://warehousewiz.com/blogs/news/warehouse-industry-statistics?srsltid=AfmBOorntQSriRYshVIJsxCqVWIXl8mTrlSqZaPXg25i8n8pXNT8b00H
- https://bhhcsafetycenter.com/warehouse-stats-facts/?print=pdf
- https://en.wikipedia.org/wiki/Warehouse
- https://mbcmusa.com/types-of-warehouse-construction/
- https://interactanalysis.com/research/warehouse-building-stock-database/
- https://www.autodesk.com/blogs/construction/cost-to-build-a-warehouse/

By utilizing additional sources of information, the validity of assumptions can be strengthened. Some parameters may be California-specific, but comparing assumptions across multiple databases lends credibility to any assumptions made, even when a California-specific database ultimately may be the source of the information. Tables showing parameters across multiple databases, and then values chosen being highlighted, can be a powerful expression of when parameters are solidly determined by being similar in multiple data sets versus being based on just a single data set. Please also note that some of the above references may include citations of relevant reports and papers that could be reviewed as further sources.

WHSC4-Prototype (A8:C8)

The area of 52,045 ft2, which is based on the CBECC 2022 model (which is based on the PNNL model), seems small given that the average from 2019 to 2022 is 226,000 ft2. The average based on the data from Dodge 2023 is not provided. The weighted average shown on the Notes Tab (B43:C59) based on Dodge 2023 would end up near 42,800 sqft. But retirements of warehouses are rapid, as they are typically located on cheap land near the boundaries between suburban and rural lands, and as suburbs expand, the land is sold for further development. The actual average size of existing warehouses is probably larger. On the other hand, CBECS shows that nationally, the warehouse and storage category averages only 17,400 sqft, but that includes buildings classified as storage, which are often smaller buildings for individual businesses' needs. Overall, for warehouses, the floor area vs the total skin area is an important factor that can drive the thermal characteristics of the building.

WHSC5- Prototype (A14:C14)

After the floor area, the other major measure impacting the skin area for a warehouse is the floor-to-floor (or in this case, the floor-to-roof) height. The assumption of 28 ft seems short. In this article from 2018:

https://www.naiop.org/research-and-publications/magazine/2018/spring-2018/business-trends/raising-the-roof/

shows that 32 (or higher) feet is probably more realistic.

WHSC6-Prototype (A15:C15)

The area above the office zone is not described. The ceiling height of the office is 14 ft, yet the building is 28 ft tall. Is the zone above the office used as part of fine or bulk storage? The zone should be identified and described.

WHSC7- Zones (n/a)

One of the critical areas of a warehouse is the loading docks. They can be open with warehouse doors away from the dock bumper (that the truck sits against) or made so that dock seals minimize the amount of air infiltration. For colder climates, the operation of the loading dock can have a large impact on the heating requirements of the building.

In addition, the goods being stored at the warehouse are usually at ambient temperature, which can mean, during cold weather, that the warehouse heating system warms the goods to the setpoint temperature, often imposing a large load on the heating system. Depending on the type of warehouse, the turnover rate for goods being stored can be quite high, with a large portion of the goods being moved in and out of the warehouse, with a storage period being as low as a few weeks. To model goods being added to the warehouse at the ambient outdoor temperature in EnergyPlus, one option is to do a side calculation of the heating or cooling rate of the mass and use a negative power level (when outdoor temperature is less than the setpoint) or positive power level (when outdoor temperature is greater than the setpoint) with OtherEquipment and a schedule with gradually decreasing (logarithmic) values. This may be automated using the EMS/PythonPlug architecture, also.

WHSC8-Zones (H2:I5)

The number of people who work in a warehouse can vary significantly. For a third-party logistics company, brands ship their inventory to the warehouse, but orders can be picked, packed, and shipped individually to stores or consumers. This may require many more people to be in the warehouse than indicated. In addition, the use of automated robots to help with picking and moving goods within a warehouse is increasing, which represents greater internal loading assumptions.

WHSC9-Zones (B2:B5)

The types of zones shown may not represent a typical warehouse well, which often consists of 5 areas:

- Loading and unloading area - This is the area where goods are unloaded and loaded into the truck. It could be part of the building or separated from the building.
- Reception area - Also known as the staging area, this is a place where the incoming goods are processed, reviewed, and sorted before proceeding to storage.
- Storage area - This is the area of the warehouse where goods sit as they await dispatch. This area can be further subdivided into static storage for goods that will take longer before being dispatched and dynamic storage for goods that sell faster.
- Picking area - This is the area where goods being dispatched are prepared or modified before being shipped.
- Shipping area - Once goods have been prepared, they proceed to the packing or shipping area, where they await the actual shipping.

https://en.wikipedia.org/wiki/Warehouse

Compared to the Office, Fine Storage, and Bulk Storage zones in the prototype, these areas have obvious differences in their occupancy and internal load characteristics.

WHSC10-Envelope (C3:D7)

The use of Gypsum board as an interior surface in warehouses seems very unlikely. Warehouses want the interior surface to be more durable to withstand impacts from moving pallets of heavy goods by forklift. While metal construction is common for warehouses, it is probably most likely to use either another interior metal panel or PVC panel, or sometimes with foam board insulation with no interior surface at all.

https://www.buildingsguide.com/build/how-to-build-warehouse/

https://www.buildingsguide.com/build/metal-building-insulation/

WHSC11-Envelope (various)

Individual rows were checked on this tab:

- Metal building U-Factor from Title 24-2025 – cells F7:X7 ok
- Metal roof decking U-Factor from Title 24-2025 – cells F12:X12 ok
- Fixed window U-Factor from Title 24-2025 – cells F17:X17 ok
- Fixed window SHGC from Title 24-2025 – cells F22:X22 ok
- Skylights U-Factor from Title 24-2025 cells F27:X27 ok
- Metal building U-Factor from Title 24-2013 – cells F6:X6 ok
- Metal roof decking U-Factor from Title 24-2013 – cells F11:X11 ok
- Fixed window U-Factor from Title 24-2013 – cells F16:X16 ok
- Fixed window SHGC from Title 24-2013 – cells F21:X21 ok
- Skylights U-Factor from Title 24-2013 cells F26:X26 ok

I could not find the Title 24-2001 version online, so those rows could not be double-checked. Continuing:

- Skylights U-Factor from Title 24-1992 cells F23:X24 ok
- Skylights SHGC from Title 24-1992 cells F28:X29 ok

The “Note” column value of “From 1992 code cycle” is redundant here since the “Selected T24” column already says “1992”, and instead the note should say the table number “Table No. 1-I” like the other rows.

WHSC12-Envelope (F13:X14, F18:X19)

Fixed window U-Factor for pre 1978 and 1978-1997 shows the note: “ASHRAE Handbook of Fundamentals for single pane & T24 2013 for double pane (50%-50%)” and uses the value of 0.74 for the U-Factor which would mean 0.112 from the ASHRAE Handbook of Fundamentals because the other value of 0.36 is shown in the table a few rows down. In the ASHRAE Fundamentals 2025, the value is shown for the Fixed column of Aluminum Without Thermal Break.

Table 4 U-Factors for Various Fenestration Products in Btu/h · ft² · °F

Product Type		Glass Only		Vertical Installation									
				Operable (including sliding and swinging glass doors)					Fixed				
Frame Type		Center of Glass	Edge of Glass	Aluminum Without Thermal Break	Aluminum with Thermal Break	Reinforced Vinyl/ Aluminum Clad Wood	Wood/ Vinyl	Insulated Fiberglass/ Vinyl	Aluminum Without Thermal Break	Aluminum with Thermal Break	Reinforced Vinyl/ Aluminum Clad Wood	Wood/ Vinyl	Insulated Fiberglass/ Vinyl
ID	Glazing Type												
	Single Glazing												
1	1/8 in. glass	1.04	1.04	1.23	1.07	0.93	0.91	0.85	1.12	1.07	0.98	0.98	1.04
2	1/4 in. acrylic/polycarbonate	0.88	0.88	1.10	0.94	0.81	0.80	0.74	0.98	0.92	0.84	0.84	0.88
3	1/8 in. acrylic/polycarbonate	0.96	0.96	1.17	1.01	0.87	0.86	0.79	1.05	0.99	0.91	0.91	0.96

Assuming that is the source, it should be more specifically documented. Obviously, the impact of fenestration on a warehouse is minimal, so this averaging 50%-50% between the two types is not going to change the answers much anyway, but the resulting U-factor of 0.74 doesn’t really exist as a product.

When this same approach is used with SHGC for pre-1978 and 1978-1997, the value shown in the table is 0.53, and the note is the same. Presumably, this may be based on the following table from the ASHRAE Handbook of Fundamentals:

Table 10 Visible Transmittance T_v, Solar Heat Gain Coefficient (SHGC), Solar Transmittance T, Front Reflectance R^f, Back Reflectance R^b, and Layer Absorptance A_n^f for Glazing and Window Systems

Glazing System					Center-of-Glazing Properties							Total Window SHGC at Normal Incidence				Total Window T_v at Normal Incidence			
					Incidence Angles							Aluminum		Other Frames		Aluminum		Other Frames	
ID	Glass Thick., in.	Center Glazing T_v			Normal 0.00	40.00	50.00	60.00	70.00	80.00	Hemis., Diffuse	Operable	Fixed	Operable	Fixed	Operable	Fixed	Operable	Fixed
Uncoated Single Glazing																			
1a	1/8 CLR	0.90	SHGC		0.86	0.84	0.82	0.78	0.67	0.42	0.78	0.78	0.79	0.70	0.76	0.80	0.81	0.72	0.79
			T		0.83	0.82	0.80	0.75	0.64	0.39	0.75								
			R^f		0.08	0.08	0.10	0.14	0.25	0.51	0.14								
			R^b		0.08	0.08	0.10	0.14	0.25	0.51	0.14								
			A_1^f		0.09	0.10	0.10	0.11	0.11	0.11	0.10								

Which shows a total window SHGC for fixed aluminum to be 0.79 when averaged with the Title 24-2013 value of 0.25, the result is 0.52, not the 0.53 shown in the scorecard rows F18:X19. A minor difference, but a bit confusing as to the source of the value. Better documentation would help here.

Another option is just to use the Title 24-1992 values for U-Factor and SHGC for pre-1978 and 1978-1997, like was done for skylights, as shown below:

TABLE NO. 1-I

PRESCRIPTIVE ENVELOPE CRITERIA FOR NONRESIDENTIAL BUILDINGS (EXCEPT HIGH-RISE RESIDENTIAL BUILDINGS AND GUEST ROOMS OF HOTEL/MOTEL BUILDINGS)

	Climate Zones				
	1,16	2-5	6-10	11-13	14-15
Windows					
U-value	0.72	1.23	1.23	0.72	0.72
Relative Solar Heat Gain					
North	0.88	0.94	0.94	0.88	0.88
Non-North	0.57	0.71	0.71	0.57	0.57

WHSC13-Envelope (F3:X3)

The metal building u-factor for pre-1978 is shown as 0.292 in all climate zones according to the note: “ASHRAE 90.1, 2022 Table A3.4.3.1, 3.5”, R-0” but the referenced table is for “Table A3.4.3.1 Assembly U-Factors for Wood-Frame Walls” and not metal building walls. Metal building walls are covered in ASHRAE 90.1-2022 Table A3.2.3, so the value should really be 1.180 instead of 0.292. See the table reproduced below:

Table A3.2.3 Assembly U-Factors for Metal Building Walls

Insulation System	Rated R-Value of Insulation	Overall U-Factor for Entire Base Wall Assembly	Overall U-Factor for Assembly of Base Wall Plus Continuous Insulation (Uninterrupted by Framing)								
			R-6.5	R-9.8	R-13	R-15.8	R-19	R-22.1	R-25	R-32	R-38
Continuous insulation only	R-0	1.180	0.136	0.094	0.072	0.060	0.050	0.044	0.039	0.030	0.026
Single compressed layer	R-10	0.186	0.084	0.066	0.054	0.047	0.041	0.036	0.033	0.027	0.023
	R-11	0.185	0.084	0.066	0.054	0.047	0.041	0.036	0.033	0.027	0.023
	R-13	0.162	0.079	0.063	0.052	0.046	0.040	0.035	0.032	0.026	0.023
	R-16	0.155	0.077	0.062	0.051	0.045	0.039	0.035	0.032	0.026	0.022
	R-19	0.147	0.075	0.060	0.050	0.044	0.039	0.035	0.031	0.026	0.022
Single layer in cavity	R-25[a]	0.059	0.044	0.039	0.035	0.032	0.029	0.027	0.025	0.021	0.019
	R-30[b]	0.052	0.042	0.037	0.033	0.031	0.028	0.026	0.024	0.021	0.019
Double layer	R-25 + R-10	0.047	0.038	0.034	0.031	0.028	0.026	0.024	0.023	0.020	0.018
	R-25 + R-16	0.042	0.036	0.032	0.029	0.027	0.025	0.023	0.022	0.019	0.018
	R-25 + R-10[c]	0.039	0.032	0.029	0.027	0.025	0.023	0.022	0.021	0.018	0.017
	R-30 + R-16	0.039	0.036	0.032	0.029	0.027	0.025	0.023	0.022	0.019	0.017

(Multiple *R-values* are listed in order from inside to outside.)
a. A minimum R-0.375 thermal spacer block or thermal break strip is required when installed without *continuous insulation*.
b. A minimum R-0.75 thermal spacer block or thermal break strip is required when installed without *continuous insulation*.
c. A minimum R-3 thermal spacer block is required.

But the assumption of no insulation in a warehouse wall because it might not have been required by an energy code is probably not valid for colder climate zones, where the goods being stored are expected to be kept above freezing temperatures. For any climate zone that has design day temperatures below freezing, I would recommend that a minimum amount of insulation, such as R-6, has likely been used even before 1978.

This probably also impacts F4:X4.

WHSC14-Envelope (E34:X40)

The infiltration I_75 values of 1.1 for Pre 1978, 1978-1997, and 1998-2007 are ok, but the 2008-2018 and 2019+ values of 1.0 and 0.6, respectively, do not appear in the referenced location of Table 140.3-B. That table does indicate that some climate zones are required to have an air barrier, which should correspond to a value from the CASE report of 0.7. But instead, a value of 1.0 or 0.6 is shown for all zones. It is not clear how the values in the scorecard were determined.

Note there is a typo in the header, and it should be “6-sided”

WHSC15-Envelope (E42)

The document that appears to be linked, titled “Infiltration_rate_report_14May2025-Short-Version.docx,” when clicked on, shows that the address is invalid. It is possible that this document would explain the information assumptions from the previous comment.

WHSC16-HVAC System (F2:F8) and Ventilation (A5:K5)

While “exhaust fan” is shown in the “other systems” column, they are not described further. The roof or sidewall exhaust fans, plus the opening of dock doors, intake louvers, and roof gravity hood intakes, comprise the entire “cooling system” of most warehouses (including the bulk storage for the warehouse prototype) that do not have mechanical cooling systems. The capacity of these exhaust fans is usually quite high and is often manually controlled to regulate temperature during warmer weather conditions. This is not described and may not be present in the prototypes. Precooling strategies are commonly used in warehouses when outdoor ventilation during the night prior to occupancy is used to cool the building. Precooling, when used, may not use a sophisticated control system but is just as likely to use manual or timer-based controls.

In addition, the use of forklifts and other motorized moving equipment may necessitate the use of much higher ventilation rates to remove exhaust fumes. For warehouses that do have higher ventilation rates due to the operation of fueled forklifts, they will often have heat recovery ventilators to minimize the energy waste during the winter months. This may be employed even if not required by code.

The use of more exhaust fans and various ventilation strategies to moderate temperatures in a warehouse impacts both the fan power and schedules and controls for the prototype model. One thing to examine in the current prototype model is when the interior temperature is above the outside temperature when the outside temperature is within the comfort range. Ventilation strategies should be added to the model to minimize when this occurs.

WHSC17-HVAC System (F2:F8)

The use of destratification fans, either large-diameter fans or higher velocity downward flow fans, is commonly used to circulate the air within a warehouse. These increase the comfort

of occupants and maintain a more consistent temperature within the storage spaces. The power required for these should be captured as another fan power item.

WHSC18-HVAC System (A2:G8) and Thermostat (A7:S10)

While “fine storage” is not really defined, it is shown as cooled and is very likely to be a controlled-humidity environment. Based on this, the control is probably not just based on temperature controls but also on humidity.

WHSC19-HVAC System (F3 and F6)

The office area is only 2,500 square feet and may be below the requirement for an air-side economizer for new construction. It is also unlikely that existing warehouses of older vintages with that size office area would have an air-side economizer.

WHSC20-HVAC System (A2:G8)

Radiant heating systems are another common heating system that is missing from the description, and are commonly used in dock loading areas of warehouses with high infiltration rates. These systems are also common in other locations that may have workers in a specific area for extended periods, such as order packaging.

WHSC21-HVAC System (A12:G18)

It would be good to add columns to this table indicating the conversion used for the inputs needed in EnergyPlus for heating and cooling efficiencies.

WHSC22-HVAC System (A22:H35)

Fan power calculations were not reviewed.

WHSC23-Thermostat (A7:S10)

See WHSC18.

WHSC24-Ventilation (A5:K5)

See WHSC16.

WHSC25-Interior Lights

Why are there not multiple vintages for interior lighting power, like there are for envelope? It seems unlikely that lighting systems get replaced often in warehouses. I would expect warehouses from pre-1978, 1978-1997, 1998-2007 to have especially poor efficacy lighting, beginning pre-LED.

WHSC26-Exterior Lights

Why are there not multiple vintages for exterior lighting power, like there are for envelope? See WHSC25.

WHSC27-Equipment (A3:C5)

A bottom-up analysis of the forklifts, sorting, and moving equipment in a warehouse may justify different values than shown. Also, I would expect fine storage and bulk storage to have different plug-load power densities.

WHSC28-Equipment (A4:C5)

For a third-party logistics company, brands ship their inventory to the warehouse, but orders can be picked, packed, and shipped individually to stores or consumers. The use of automated robots to help with picking and moving goods within a warehouse is increasing, which represents greater internal loading assumptions.

WHSC29-Schedules (A19:AI21, A38:AI40)

A decrease in infiltration for a warehouse during occupied hours seems unlikely, given that it is when dock doors are more likely to be open. These assumed schedules look like ones for a pressurized building, such as an office building, which is extremely unlikely for a warehouse using exhaust fans with gravity dampers. The small office area is also unlikely to

have decreased infiltration during the day since the door from the office to the warehouse is frequently (or maybe continuously) open.

WHSC30-Schedules (A4:AI6)

The Saturday occupancy schedule seems unlikely. Generally, warehouses are simply open or closed and fully staffed when open. Saturday being open and fully staffed seems more likely than Saturday having only 20% occupancy and shorter hours. If this is changed, it would impact many other schedules.

WHSC31-Thermostat (A11:S11) and HVAC System (B5:F5, B8:F8)

Allowing only heating only systems for bulk storage is likely to be violating a new law that requires employees to be in conditioned space. Title 8, Division 1, Chapter 4 adds a new section 3396 titled "Heal Illness Prevention in Indoor Places of Employment" which is very likely to impact new and existing warehouses.

https://www.dir.ca.gov/oshsb/documents/Indoor-Heat-updated-txtbrdconsider.pdf

The law essentially requires air conditioning anywhere that employees may face warm temperatures.

WHSC32-Thermostat (B:6, B10)

The second lines in each group that refer to "cooling setpoint" should probably be called "cooling setback" similar to how "heating setback" is mentioned. For cooling this is not always referred to be "cooling setback" but some way to differentiate unoccupied set point is needed.

WHSC33-Equipment (A3:C5)

Electric battery forklifts should be assumed for all new warehouses and a transition to them for existing warehouses should also be assumed based on the typical life of a propane fueled forklift. This is based on California Air Resources Board requirement for Zero-Emissions Forklifts:

https://ww2.arb.ca.gov/our-work/programs/zero-emission-forklifts

This impacts internal loads since charging of batteries for the forklifts is a significant load.